

BTF Trace Viewer

Version 1.4.0 — Desktop (Python) and Web

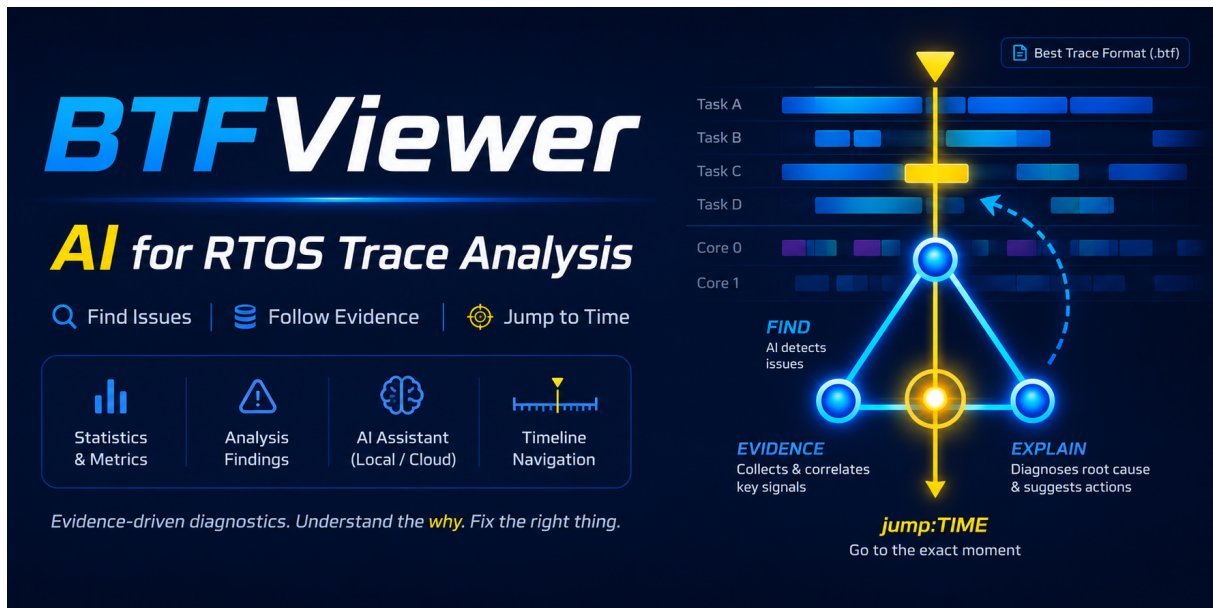


Figure 1: BTFViewer AI-assisted

BTFViewer is an **AI-assistant tool for RTOS trace analysis**: find evidence in a captured schedule, then explain it. Open FreeRTOS context-switch traces in **Best Trace Format** (.btf), inspect the timeline, measure with cursors, review statistics, and ask the **AI** tab to cite findings and walk the cause.

BTF Trace Viewer

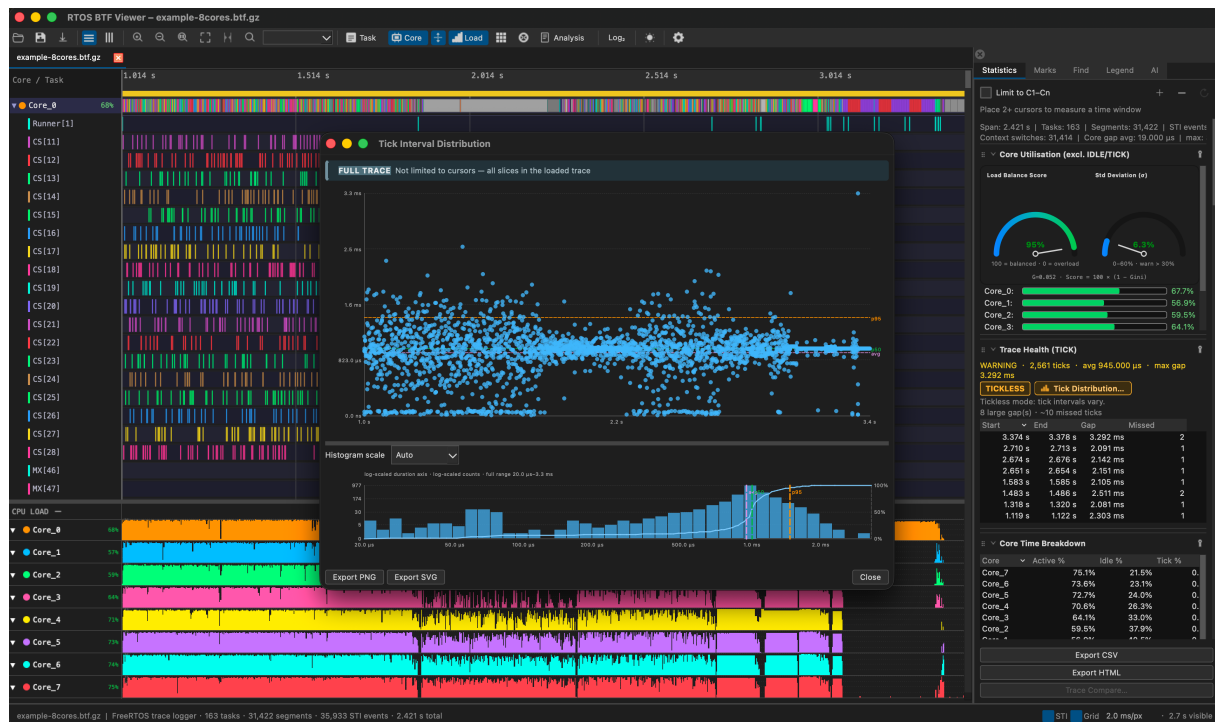


Figure 2: BTF Viewer screenshot

[Try the live demo](#)

Heading level: ■ section · ■ subsection · ■ topic · ■ detail

Features ■

- **Timeline visualisation** — task and core views, horizontal or vertical layout, smooth zoom and pan
- **Multi-trace sessions** — open several traces in tabs; compare builds side by side
- **Measurement tools** — cursors, bookmarks, annotations, find & jump
- **Statistics & analysis** — utilisation, latency, migrations, sync/mutex metrics, and severity-tagged Analysis Findings

- **AI Assistant** — find evidence and explain Analysis Findings (import examples/ai presets; local or cloud)
- **Export** — annotated snapshots (PNG/SVG), CSV/HTML reports, Perfetto traces
- **Desktop CLI** — headless reports and images for scripts and CI
- **Desktop and Web** — the same analysis workflow in a desktop app or the browser
- **Scripted demo** — 8-core walkthrough with narration ([Demo](#))

Metric definitions and formulas are in [Statistics](#). Step-by-step diagnosis playbooks: [WORKFLOWS.md](#).

Contents

Documentation map

Document	Primary question	Responsibility
README.md	How do I use BTFViewer?	Product overview: AI assistant for RTOS traces, installation, timeline, analysis, export
WORKFLOWS.md	How do I diagnose a problem?	Step-by-step analysis and investigation playbooks
STATISTICS.md	What does this measurement mean?	Deterministic BTF metrics, distributions, scope, comparison, interpretation
AI.md	How does AI-assisted investigation work?	AI architecture, models, tools, privacy, CLI, evaluation, implementation

Recommended path:



Use **README** for the product (find evidence and explain), **WORKFLOWS** when you have a symptom to investigate, **STATISTICS** when you need metric-level detail, and **AI** when you need AI configuration or implementation details.

Section

Quick start	Install and open a trace
Demo	Scripted tour, voice packs, recording
Using the viewer	Toolbar, navigate, measure, and mark
Analysis	Findings, BTF analysis pages, Trace Compare
AI Assistant	Investigation, evidence, workflows, privacy
Statistics	Metric meaning, formulas, charts, how to diagnose
Export	Snapshots, reports, Perfetto, CLI
Settings	Preferences and defaults
Keyboard & mouse	Shortcuts reference
Further reading	Workflows, AI details, developer notes

Quick start

Desktop

Requirements: Python 3.8+ ([PySide6](#) ≥ 6.4). Python 3.9+ is recommended.

BTF Trace Viewer

```
cd BTFViewer
pip install -r requirements.txt
python builds/btf_viewer.py [trace.btf]
```

With no file argument, the previous session is restored. You can also use **File → Open**, drag-and-drop, or **File → Open Recent** (Desktop).

Web

Open the standalone build in any modern browser (no server required for typical use):

```
open BTFViewer/builds/btf_viewer.html # or double-click the file
```

Or use the [hosted demo](#). Toolbar **Demo**, opening a demo .xml, and **Record** are in **Demo**. To rebuild from source:

```
cd BTFViewer && make web # → builds/btf_viewer.html
```

AI: After a trace is open, import [examples/ai/presets.json](#) in **Settings → AI** for ready-made Ollama / OpenAI / Gemini / DeepSeek / Grok endpoints. See [AI Assistant](#) (user guide) and [AI.md](#) (setup, tools, troubleshooting).

Supported files

Format	Notes
.btf	Best Trace Format
.btf.gz / .gz, .btf.bz2 / .bz2	Compressed single traces
.btf.zip / .zip	One or more .btf members — each opens in its own tab. Desktop keeps archive.zip::member.btf paths (re-read on launch). Web titles multi-member zips archive.zip::member so two archives cannot collide.

Format	Notes
.xml (W)	Demo pack script — see Demo .
.xtf	Shareable demo tour pack (zip of xml + btf + voice/) — Open or drag to play.

Sample traces are under `tracedata/` (for example `example-2cores.btf.gz`).

Demo

Scripted tour of the 8-core sample. Desktop (`make demo`) and Web (toolbar **Demo**, or **Open** the pack) share [demos/demo_8cores/](#). [Try the live demo](#). XML actions and the demo HTTP API: [demos/README.md](#).

Desktop recording

`make demo` launches `builds/btf_viewer.py` on the pack XML (rebuilds the bundle if `btf_viewer_pkg/` is newer). **Open** or drop a demo .xml, .xtf, or pack folder to play the same in-app tour as Web: demo bar, overlay pointer, and XML APIs. Moving the real mouse hides the overlay pointer until the next scripted `<move>`.

```
cd BTFViewer
make demo
make demo DEMO_LANG=zh-tw
python3 builds/btf_viewer.py demos/demo_8cores/demo_8cores.xml
python3 builds/btf_viewer.py builds/demo_8cores.xtf
```

Narration defaults to the XML `<languages default>` (**English**). Override with `DEMO_LANG`, `BTFVIEWER_DEMO_LANG`, or the demo bar **Voice** menu (remembered in `builds/btf_viewer.rc`). **Space** pauses; **Esc** twice (within 2.5 s) stops.

Engine: [btf_viewer_pkg/demo_inapp.py](#) (lockstep with Web [web/src/utils/demoRunner.js](#)).
Opt-in HTTP [demo_api.py](#) remains available as `BTFVIEWER_DEMO_API=1`.

Web pack and recording

Toolbar **Demo** loads the bundled sample without a file picker. **Open** a demo .xml, a shareable .xtf pack, or drop the pack folder to run the same script as desktop make demo. After picking the XML alone, **Open pack folder** opens a folder dialog (Select/Open is enabled on a directory). The dialog may start in Documents on macOS; navigate to the pack folder. Dropping the pack folder or a .xtf onto the viewer skips this step. The viewer then loads the pack's .btf.gz, plays voice/<lang>/*.mp3, and drives view mode, Load, Statistics sections, highlights, cursors, and Analysis through in-app APIs. <move> / <sweep> animate a pointer overlay to the XML <targets> (the browser cannot move the OS cursor; hover events still fire so timeline tooltips appear). Moving the real mouse hides the overlay pointer until the next scripted <move>. Opening a .btf still loads the trace as usual.

Build a shareable pack (choose voice packs with `--voice / DEMO_LANGS`):

```
make demo-pack                                # → builds/demo_8cores.xtf (en only)
make demo-pack DEMO_LANGS=en,zh-tw            # English + 中文
make demo-pack DEMO_LANGS=all                 # every voice/<lang>/
python3 scripts/demo_pack.py demos/demo_8cores --list-voices
python3 scripts/demo_pack.py demos/demo_8cores --voice en,zh-tw -o
↪ builds/demo_8cores.xtf
```

An .xtf is a zip of *.xml (languages filtered; audio paths → .aac), *.btf*, and voice/<lang>/ (MP3 transcoded to AAC 24 kHz mono 32 kb/s via ffmpeg). Open or drag it in **Web or Desktop** to play the tour. Use `--keep-mp3 / make demo-pack` with a custom pack command to skip AAC conversion.

Record uses the browser's display capture; choose this tab and include tab audio. The mouse pointer is included (page overlay for tab capture; native pointer for window/screen). Tab capture does not include the OS pointer, so Record paints one in the page.

The demo bar's **previous / pause / next** icons skip or hold the current section (pause also holds narration; the icon switches to play to resume). **Space** pauses and resumes. **Esc** twice (within 2.5 s) stops the script.

Voice packs

Every language uses the same layout. XML paths stay `voice/01_title.mp3`; the runner looks in `voice/<lang>/` first, then flat `voice/`, then `voice/<default>/`.

```
text/<lang>/01_title.txt
voice/<lang>/01_title.mp3
voice/<lang>/voice.json
```

Shipped languages for `demo_8cores`: **en** (English, default), **zh-tw** (中文). A **Voice** menu on the demo bar restarts the current section in that language (Web: remembered in the browser, else the browser locale; Desktop: remembered in settings, else `BTFVIEWER_DEMO_LANG / DEMO_LANG`, else the UI locale).

```
python3 scripts/demo_voice.py status demos/demo_8cores
make demo-voice
```

Install, export, render, and sync-xml: [scripts/demo_voice.py](#) and [demos/README.md](#). Write **Free-RTOS** (with a hyphen) in narration scripts so speech engines pronounce it correctly.

Pack layout

Path	Purpose
demos/demo_8cores/demo_8cores.xml	Runner script (steps / actions)
demos/demo_8cores/demo_8cores.btf.gz	Frozen trace (stable vs tracedata/)
demos/demo_8cores/text/<lang>/	Narration scripts (.txt)
demos/demo_8cores/voice/<lang>/	TTS audio + voice.json

Using the viewer ■

Desktop and Web share the same concepts. Where behaviour differs, it is noted below.

Toolbar ■

Hover any control for its name. Icons that need a loaded trace (snapshot, SVG, Perfetto, crop, Heatmap, Analysis) stay disabled until a .btf is open. **Compare** needs two or more tabs.



Figure 3: Desktop toolbar

Left to right (Desktop; Web is the same cluster except as noted):

Icon / control	Meaning
Folder	Open a BTF trace (Ctrl+O). Web also opens a demo .xml pack.
Camera	Snapshot editor — annotated PNG of the viewport (Ctrl+S)
Download	Save SVG of the current viewport (Ctrl+Shift+S)
Window / trace	Perfetto — export Chrome Trace JSON (Ctrl+Shift+E)
Crop square	Save BTF — write the cursor window C1-Cn as .btf / .btf.gz
Horizontal bars	Horizontal layout — time runs left → right
Vertical bars	Vertical layout — time runs top → bottom
Magnifier +	Zoom in (Ctrl++)
Magnifier –	Zoom out (Ctrl+-)

Icon / control	Meaning
Magnifier reset	1:1 — configured zoom density
Corners	Fit the whole trace (Ctrl+0)
Range arrows	Zoom to cursor range between earliest and latest cursor (Ctrl+R; needs 2+ cursors)
Magnifier	Find (Ctrl+F)
Fit combo	Zoom preset — 1% ... 75% of the trace, or Fit
Task	Task View — one row per task (merged across cores)
Core	Core View — one expandable row per CPU
Expand arrows	Expand / collapse all cores (Core View only)
Load	Show / hide the CPU load chart under the timeline
Grid	Heatmap — Migration & Corridor Inspector (multi-core traces)
Analysis	Analysis Findings — heuristic triage for the current Statistics scope
Overlap squares	Compare — Trace Compare between two open tabs
Log₂	STI waveform y-axis linear ↔ log ₂ (when an STI row is expanded)
Sun / moon	Theme — switch light / dark
Gear	Settings (Ctrl+,) — far right
Question mark	Help — keyboard shortcuts

Web adds **Demo** (bundled 8-core tour), **Record** (tab capture), and an **About** control; a **More** overflow holds the same actions when the window is narrow. **All tasks** appears on the toolbar only while a heatmap task filter is active.

View modes and orientation 

Control	Purpose
Task View	One row (or column) per task
Core View	One expandable row per core; expand a core or use Expand / Collapse All
Horizontal / Vertical	Time left→right or top→bottom

Switch from the toolbar (both apps) or the **View** menu (Desktop). The last orientation is remembered. On Web, 1 / 2 switch Task / Core View and H / V switch orientation.

Zoom and pan

- **Wheel** — pan (add **Ctrl** to zoom; **Shift** swaps axes)
- **Pinch** (macOS) — zoom
- **Middle-drag** — select a time range to zoom
- **Fit** (Ctrl+0 / F) — show the full trace
- **Zoom preset** — toolbar menu: **1%**, **2%**, **5%**, **10%**, **25%**, **50%**, **75%**, **Fit** (same as Desktop; percent of the full trace that is visible)
- **1:1** — reset to the configured zoom density
- **Zoom to cursor range** (Ctrl+R) — fit between the earliest and latest placed cursor (2+ cursors)

Labels, legend, and highlight

- Drag the label-column edge to resize; double-click the edge to auto-fit.
- The **Legend** lists tasks with colours and filters (including migrated tasks).
- Click or hover a task label or legend row to highlight that task on the timeline.
- Hover a segment for duration, core, and neighbouring activity.

CPU Load

Toggle **Load** on the toolbar for a utilisation chart under the timeline. Drag the divider to resize. With two or more cursors, the chart can show a cursor-range average.

With a task **lock-highlighted**, Task View shows that task's utilisation **per core** in the load strip — useful when investigating migrations (see [Highlight a migrating task](#)).

Cursors

Place up to **4-8** cursors (default 4; set in Settings).

Action	How
Place	Click the timeline, or press C at the viewport centre
Move	Drag the cursor line
Remove	Click near a cursor, or use the context menu
Clear all	Shift+C or Shift+right-click
Snap to boundary	Shift+click

With **two or more** cursors, the status bar shows a short range summary, and Statistics can limit metrics to that window (**Limit to C1-Cn**). Toolbar crop / **File → Save selection as BTF...** exports the cursor window: both apps prefer the original BTF text (desktop re-reads the file / zip member; web uses in-memory sourceText), and **reconstruct** a resume/preempt + STI subset if that source is missing (e.g. web after refresh).

Marks and find

Tool	Purpose
Bookmarks	Named timestamps (Ctrl+B)
Annotations	Free-text notes at a time (Ctrl+Shift+B / A)
Find	Search tasks, migrations, STI events, intervals, and more (Ctrl+F; F3 / Shift+F3 to step)

Find modes (match count is shown at the top of the panel; each mode has a short description under the dropdown):

Mode	Matches
Contains	Substring on task names and annotation notes
Exact	Whole-string task merge key / name
Regex	Case-insensitive regex on tasks and annotations
Migrations	Migration boundaries by task or core (Core_0, CS[22], ...)
STI	STI channel, event, note, and core
Intervals	Interval start/stop spans and interval STI notes
Lifecycle	Task create / delete / suspend / resume STI
Pointers	Mutex / semaphore / queue 0x... pointers and sync notes

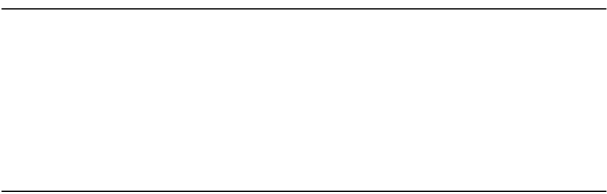
Right-click the timeline for cursor, bookmark, and annotation actions. Marks and cursors are restored with the session. Clicking a Statistics chart point adds an annotation **without switching right-panel tabs**.

Multi-tab traces

Open several files (or a multi-BTF zip) as tabs. Each tab keeps its own zoom, cursors, marks, and filters. Cycle with Ctrl+Tab / Ctrl+Shift+Tab; close with Ctrl+W. Desktop restores tabs from disk (including zip members). Web restores up to **8** packed traces from IndexedDB by tab name (private mode / quota may skip tabs; clearing site data drops them).

Tag and STI markers ■

Software-trace items appear as markers (and optional tag channels). Toggle STI rows with I or Settings. Expand tag channels for waveforms where available.



Analysis ■

Start with toolbar **Analysis** for a severity-tagged triage of the current Statistics scope, then open the named Statistics sections. Those pages are **deterministic BTF analysis** (facts first). The **AI** tab explains and navigates them — it does not replace them. For AI-assisted investigation, see **AI Assistant**. Playbooks: **WORK-FLOWS.md**.

BTF analysis pages ■

Viewer analysis stays on **BTF → statistics → visualization → comparison**. There is no source/ELF inspection, scheduler simulation, or invented kernel response time (BTF has no release/completion pair). **Apply cursors** on a finding recommends a window; it does not apply until you click.

Page	What to use it for
Timeline Anomalies	Unusual regions (spikes, bursts, idle, deadline misses) without relying only on Findings. Investigate... opens the AI template.

Page	What to use it for
Worst Events	Top-N outliers; click a row to jump. Response p99 sits next to execution Max.
Response Time	Min / mean / median / P90-P99.9 / max / σ / jitter. Click P99 like Execution.
Period / Jitter	Activation gaps, sparkline, missed / extra / burst / long-gap counts.
Unified Jitter	Execution, period, dispatch (STI), and wake (block-wait) CVs.
Distribution Explorer	Histogram for one metric $\times$ task.
Task $\times$ Core / Core Utilization Over Time	Hot vs idle cores, load imbalance, saturation windows.
Preemption Matrix / Chain / Story	Who preempted whom, duration, recurring chains.
Mutex Blocking / Waiter $\times$ Owner	Wait totals, top blockers, owner handoff.
Critical Path	Execution / preemption / mutex / migration / Other; click a component.
Task Health	Heuristic 0-100 score (not an AI probability).
Recurring Patterns	Repeat incidents in one trace; Compare lists patterns shared across traces.
Trace Compare	A vs B including Response P99, mutex, deadline misses, and a deterministic Why? .

Analysis Findings

Toolbar **Analysis** summarises likely issues for the current scope (load imbalance, WCET/CPU hotspots, blocking, priority inversion, core thrashing, deadline breaches, tick health, sync/mutex bounces, and similar). On traces with **two or more cores** and positive total utilisation, a **Core utilisation balance** finding always includes **Load Balance Score ...% (σ =...%, G =...)** — Desktop and Web match, including

when cores look **reasonably balanced** (Score $\geq 85\%$ and $\sigma \leq 30\%$). Finding ink follows dark/light theme (info is not forced to dark-on-dark). The dialog opens with an **overview**: trace-quality warnings, related findings grouped as incident clusters ([I1], [I2], ...), and a suggested phase window. Finding times are drawn as dashed overlays on the timeline. From the dialog, left to right: **Query with AI...** walks the findings card, **Investigate...** runs an evidence-driven drill-down (tools + cursors), **Verify with AI...** checks the selected finding, **Explain...** / **Root cause...** go deeper, **Auto investigate...** runs the full tool chain, **Save recipe...** stores a user investigation template, **Story...** exports the overview plus findings, **Apply cursors** places C1-C2 on a recommended window for the selected finding, or **Save as Text...** for a copy. The same card appears in **Export HTML**. **Ctrl+K** jumps to Analysis, workspace presets (Triage / Latency / SMP / Compare), Inspect task, and other surfaces without extra toolbar buttons. The status bar inspector shows the pinned task and the first trace-quality warning.

How to act on a finding

1. Note the severity and the Statistics section it names.
2. Open that section, sort by Max / Rate / Bounce as relevant.
3. Click **Min** / **Max** / **p95** / **p99**, a chart point, or an inspector cell to jump the timeline. Or click **Apply cursors** on the selected finding.
4. Place cursors around the phase of interest and enable **Limit to C1-Cn**.
5. Optionally click **Query with AI...** / **Investigate...** / **Verify with AI...** / **Explain...** / **Root cause...** / **Auto investigate...** (or open the **AI** tab) ([WORK-FLows.md §7](#)).

How to find problems (quick map)

Symptom	Start here	Then check
Unknown — triage first	Toolbar Analysis	Named Statistics sections
Tick jitter / tickless	Trace Health (TICK)	Scope a busy window; Execution Max
SMP uneven load	Core Utilisation (Score / σ)	Concurrent Core Active; Migrations

Symptom	Start here	Then check
High scheduler cost	Kernel Switch Overhead	Core Time Breakdown (Gap %)
Task too slow on CPU	Execution Time (Max / p95)	Preemption Chain, Mutex
Task waits too long	Blocking Time	Preemption Chain, Mutex
Ready→run delay	Dispatch Latency	Blocking, Preemption (needs STI resume)
Priority inversion	Priority Inheritance	Mutex pairing, Blocking
Core thrashing	Core Migrations (Rate, Ping)	Heatmap / Chord; Mutex Bounces
Lock / queue issues	Mutex / Semaphore / Queue	Blocking, Migrations
Before/after change	Trace Compare	Same cursor phases on both tabs

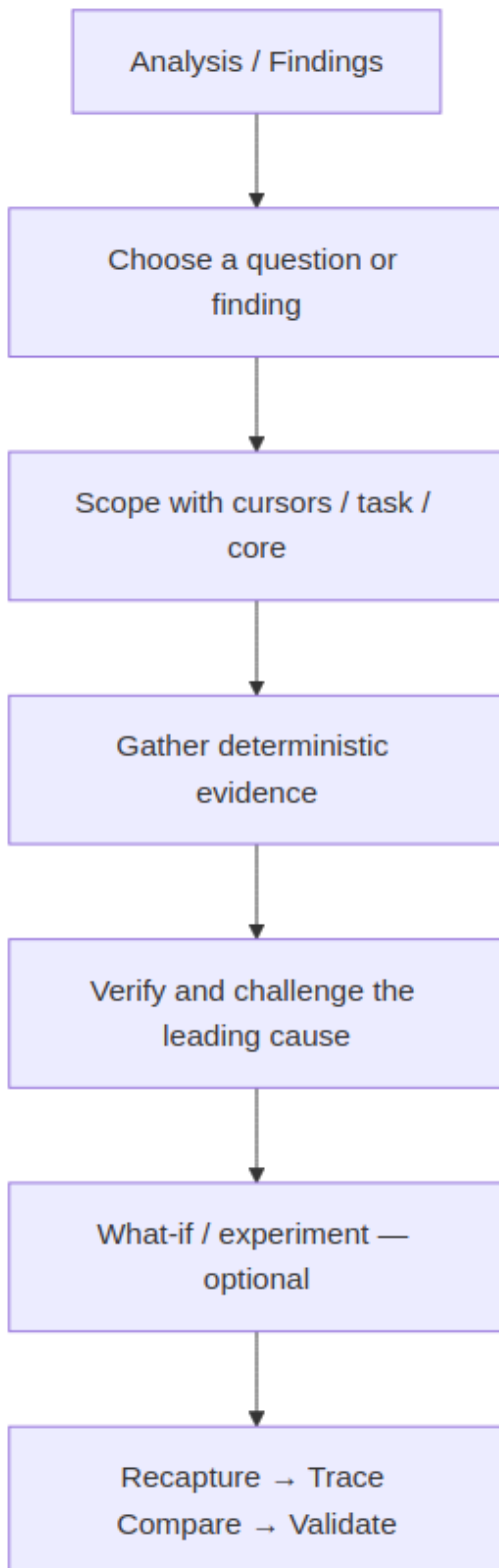
Trace Compare

With **two or more** tabs open, toolbar **Compare** (right after **Analysis**) diffs summary, top tasks, utilisation, migrations, execution, blocking, inter-arrival, preemption, sync, **Response P99**, and **mutex blocking**. The **Trends** page lists every open tab (3+) with span, migrations, load-balance, and tick health. The summary strip includes those latency/blocking deltas plus **deadline-miss** counts from **Settings → Display** task deadlines, a deterministic **Why?**, and shared recurring patterns. Optionally limit each side to its own cursor range. **Save as baseline / Score vs baseline** store and z-score Trace A metrics (same profile as `baseline_score`). Export CSV/HTML from the dialog, **Validate experiment...** to score expected vs actual deltas in the **AI** tab (`validate_experiment`; actual percents come from this compare, including **Scope to cursors**), or **Query with AI...** to walk the same tables (Trace Compare template). See also [Core migration analysis](#).

AI Assistant

BTFViewer is an **AI-assistant tool for RTOS trace analysis**. The **AI** tab **finds evidence** in Analysis Findings and Statistics and **explains** it. It does not replace the timeline or invent measurements from the raw .btf.

Recommended workflow



Goal	Start here	Detailed guide
Triage a trace	Analysis → Query with AI...	WORKFLOWS.md
Explain or verify a finding	Select a finding → Explain... / Verify with AI...	AI.md
Investigate a time region	Place ≥2 cursors → Explain this region with AI	WORKFLOWS.md
Compare before / after	Trace Compare → Query with AI...	WORKFLOWS.md
Test a proposed change	What-if / Optimize → recapture → compare	AI.md
Configure models / endpoints	Settings → AI	AI.md

Important: do not request mitigations before the timeline and Statistics agree with the finding. AI responses are constrained by available evidence and include validation / confidence information where applicable.

See [AI.md](#) for models, endpoints, GUI tools, privacy, CLI, benchmark methodology, planner, causal engines, and implementation details. See [WORKFLOWS.md](#) for repeatable analysis and investigation playbooks.

Same panel on **Desktop** and **Web**. Findings can include WCET **Max»Avg** spikes. Optional experiments use `what_if / optimize_experiment` (**Heuristic slice-replay, not FreeRTOS kernel**; Ranked `optimize_experiment`).

Product entry points stay here; the system reference is [AI.md](#). The AI tab shows a Triage → Scope → Investigate → Verify → Experiment → Compare stepper (click a stage to jump in the log). **Start Investigation** (empty log) runs **Auto investigate**. Restart restores an in-progress case only when the log still has a user or assistant turn; otherwise **Start Investigation** stays available and a leftover **Current Issue** card is not restored. **Clear** removes chat replies, resets the usage meter, and clears current investigation issues. The usage bar shows **Context: Compact · 4.6k tok · 3 tools · 12s**. **Settings** → **AI** → **Context** is Compact / Balanced (default) / Full evidence. What-if stays on **Verify** until verify tools or strong evidence quality; Experiment is labeled as a heuristic estimate (recapture and Compare to measure).

Triage findings, Verify finding, Auto investigate, and the **Investigation plan** checklist live in the same tab. **Cheapest evidence first** (planner). Keys: Settings → AI → API key first, then OPENAI_API_KEY / GEMINI_API_KEY / OLLAMA_API_KEY (OPENAI_API_KEY, then GEMINI_API_KEY, then OLLAMA_API_KEY). Local Ollama needs none. Web can inject the same names via window.__BTF_AI_ENV__. CURSOR_API_KEY is for live ai-test XML <api-key env="VAR"> and is ignored for chat / Test connection. CLI: [AI.md](#) → [CLI regression gate](#).

Statistics

Statistics are the deterministic measurement layer of BTFViewer. Use them to quantify execution, blocking, jitter, scheduling, synchronization, migration, and comparison behavior.

For the complete metric definitions, formulas, distributions, scope rules, and interpretation guidance, see **STATISTICS.md**.

Quick map

Symptom	Start here
Tick jitter / tickless	Trace Health (TICK)
SMP imbalance	Core Utilisation
Task too slow	Execution Time
Task waits too long	Blocking Time
Ready → running delay	Dispatch / Scheduling Latency
Priority inversion	Priority Inheritance
Core thrashing	Core Migrations
Lock / queue problems	Mutex / Semaphore / Queue
Before / after change	Trace Compare

Export

Action	Desktop	Web
Annotated snapshot	Toolbar snapshot / File → Snapshot Editor...	Toolbar snapshot
Copy viewport	Clipboard	Clipboard
SVG	Toolbar / File → Save SVG	Toolbar Save SVG
Perfetto JSON	Toolbar / File → Export Perfetto...	Toolbar Perfetto
Cursor-range .btf	Toolbar crop / File → Save selection as BTF... (C1-Cn; .btf / .btf.gz; filter source, else reconstruct)	Toolbar crop (2+ cursors; .btf download; same filter / reconstruct)
Statistics CSV/HTML	Statistics panel	Statistics panel
Compare CSV/HTML	Trace Compare dialog	Trace Compare dialog

Headless CLI (Desktop only)

Same engine as the GUI, suitable for CI. Use `QT_QPA_PLATFORM=offscreen` when no display is available.

Command	Purpose
info	Trace summary (--json optional)
report	Full statistics CSV/HTML
compare	Two-trace diff (two paths or one multi-BTF zip)
analyze	CI regression gate vs baseline .btf or metrics JSON (--fail-on-regression; optional --ai; --save-baseline) — details in AI.md
ai-test	AI evidence/validator benchmark (tests/ai; live --config examples/ai/benchmark.xml, optional --compare-context for Compact/Balanced/Full, -o AI_BENCHMARK.md) — AI.md

BTF Trace Viewer

Command	Purpose
migrations	Migrations table as CSV
snapshot	PNG/SVG of timeline, migration inspector, or a metric plot
perfetto	Chrome Trace JSON
slice	Timestamp window as a smaller .btf

```
python builds/btf_viewer.py info trace.btf
python builds/btf_viewer.py report trace.btf -o report.html --format html
python builds/btf_viewer.py compare before.btf after.btf -o diff.html
python builds/btf_viewer.py analyze candidate.btf --baseline baseline.btf
  ↪ --fail-on-regression
python builds/btf_viewer.py analyze candidate.btf --save-baseline
  ↪ /tmp/base.json
python builds/btf_viewer.py snapshot trace.btf -o view.png --view timeline
python builds/btf_viewer.py perfetto trace.btf -o trace.json
python builds/btf_viewer.py slice trace.btf -o window.btf --lo 100000 --hi
  ↪ 500000
```

Run `python builds/btf_viewer.py <command> -h` for full options.

Settings

Open **Settings** from the toolbar or `Ctrl+,`. Toolbar **Help** opens the keyboard short-cut list (Web: also ?).

Area	What you configure
Appearance	Dark/light theme, fonts, colorblind-safe palette. Desktop font sizes are pt (HiDPI-scaled); web sizes are CSS px . Defaults look similar; the numbers are not interchangeable.

Area	What you configure	
Display	Show/hide Legend, Statistics, Marks, Find, AI, CPU Load; Timeline overlays (STI, grid, hover highlight); Analysis thresholds (CPU budget % and per-task deadline ns)	
Layout	Label width, row height, zoom 1:1 density, max cursors, time decimals, CPU/STI sizes	
AI	Enable, Auto-apply GUI actions , Context (Compact / Balanced / Full evidence), Anonymize task names for cloud , Treat this trace as sensitive , Log MCP messages to file (Desktop debug; off by default), preset (Ollama / OpenAI / Gemini / Custom), base URL, model, authentication (none / API key / Sign in), Allow self-signed TLS (Desktop only; browsers still verify certificates), reply language	
	Desktop	Web
Storage	bt _f _viewer.rc next to the viewer	Browser localStorage
Apply	Live preview while the dialog is open; OK persists; Cancel reverts	Same (OK persists; Cancel reverts)

Sessions also restore open tabs, viewport, cursors, and marks. Desktop re-opens filesystem paths (including archive.zip::member.bt_f). Web restores tab names plus IndexedDB packed traces (max 8). **Reset to Defaults** restores compiled defaults (theme, layout, AI form, and Statistics pins / order / expand-collapse) immediately in the viewer; **OK** writes them to storage, **Cancel** undoes the preview.

Keyboard & mouse

Shortcuts marked **(W)** are Web-only. Others work on Desktop and Web. On Web, press ? for an on-screen cheat sheet.

File & tabs

Key	Action
Ctrl+O	Open file (new tab)
Ctrl+W	Close tab
Ctrl+Tab / Ctrl+Shift+Tab	Next / previous tab
Ctrl+Q	Quit (Desktop)
? (W)	On-screen shortcut cheat sheet

View & zoom

Key	Action
Ctrl++ / Ctrl+-	Zoom in / out
+ / - (W)	Zoom in / out (no modifier)
Ctrl+0 / F	Fit to window
Ctrl+R	Zoom to cursor range (earliest-latest cursor)
Ctrl+,	Settings
Ctrl+K	Command palette (Analysis, Statistics, AI, Compare, workspace presets, Inspect task, ...)
G / I / D	Grid / STI / theme
1 / 2 (W)	Task View / Core View

BTF Trace Viewer

Key	Action
H / V (W)	Horizontal / vertical layout
Ctrl+G	Jump to time
Ctrl+Home / Ctrl+End	Trace start / end

Edit & marks

Key	Action
Ctrl+Z / Ctrl+Y	Undo / redo
C / Shift+C	Place / clear cursors
B / M / Ctrl+B	Bookmark
Shift+B	Clear all bookmarks
A / Ctrl+Shift+B	Annotation
Shift+A	Clear all annotations
Tab / Shift+Tab	Next / previous segment
← → ↑ ↓	Pan (Shift+arrow: prev/next boundary)
Ctrl+F / F3 / Shift+F3	Find / next / previous

Export

Key	Action
Ctrl+S / S (W)	Snapshot editor
Ctrl+Shift+C	Copy viewport

BTF Trace Viewer

Key	Action
Ctrl+Shift+S	Save SVG
Ctrl+Shift+E	Export Perfetto

Mouse

Action	Effect
Wheel / Ctrl+wheel	Pan / zoom
Left-drag background	Pan
Middle-drag	Zoom to selection
Ctrl+left-drag	Measure time between two points (double-arrow ruler + Δ time; release Ctrl or button to hide)
Click timeline	Place cursor
Drag cursor or mark	Move
Right-click	Context menu

Further reading

Document	Audience
WORKFLOWS.md	Analysis playbooks and AI ask order

Document	Audience
AI.md	AI architecture, tools, models, planner, troubleshooting, CLI
STATISTICS.md	Detailed metric definitions, formulas, charts, and analysis reference
btf-viewer-slides.md	Presentation overview
demos/README.md	Demo XML actions, HTTP API, voice-pack tooling

Developer notes

Day-to-day users can ignore this section.

Task	Command
Rebuild Desktop + Web	<code>make -C BTFViewer</code>
Desktop package only	<code>make -C BTFViewer bundle → builds/btf_viewer.py</code>
Web only	<code>make -C BTFViewer web → builds/btf_viewer.html</code>
Guided demo	<code>make -C BTFViewer demo</code> — see Demo
Tests	<code>make -C BTFViewer test (desktop) / test-web / test-all / ai-test / ai-test-live / ai-test-context</code>
Docs PDF	<code>make -C BTFViewer doc → builds/{README,STATISTICS,AI,WORKFLOWS,btf-viewer-slides}.pdf</code> (Mermaid via Node/npx @mermaid-js/mermaid-cli)
Dev run (Desktop)	<code>python -m btf_viewer_pkg [trace.btf] from BTFViewer/</code>

Edit sources under `btf_viewer_pkg/` and `web/`; commit regenerated files under `builds/` with your changes. Keep AI tool schemas and mermaid layout in sync

(ai_tools.py / ai_mermaid.py ↔ web/src/utils/aiTools.js / aiMermaid.js; mermaid_palette(is_dark) ↔ mermaidPalette(dark)). Parser and Statistics numbers are pinned by shared goldens (tests/fixtures/*-golden.json) asserted from both tests/test_parser_golden.py / tests/test_stats_web_parity.py and web/tests/. Synthetic traces: scripts/gen_trace.py --help. BTF field reference: [TRACE_FORMAT.md](#). Live suite XML: [AI.md](#) → [Benchmark / evaluation suite](#). Recorded scores: [AI_BENCHMARK.md](#).

Contributors

Thanks to everyone who has contributed to this project.

Contributor	Contribution
DiogoRoseira	CPU Load Graph and metrics distribution charts

Documentation navigation

Document	Question answered
README.md	How do I use BTFViewer?
WORKFLOWS.md	How do I diagnose a problem?
STATISTICS.md	What does this measurement mean?
AI.md	How does AI-assisted investigation work?
